REVIEW–1 DOCUMENTATION

AI Assisted Data Visualization for Business Decision Support Systems

Engineering Capstone Project – 1
A.Y. 2026–2027

Project Domain	Business Analytics and Decision Support Systems
Review	Review–1
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1. Project Title

AI Assisted Data Visualization for Business Decision Support Systems

2. Problem Definition

The rapid growth of business data creates a need for systems that can transform raw transactional information into meaningful insights for effective decision-making. Conventional analysis can require substantial manual data preparation, exploratory analysis, model comparison, and interpretation. The proposed project addresses this challenge by developing an AI-assisted analytics platform that combines data science, machine learning, explainable AI, and interactive visualization to support business decision-making.

The project focuses on business datasets containing information such as sales, profit, customers, products, discounts, regions, orders, shipping, payments, deliveries, and reviews. The intended system connects the analytics pipeline to an interactive dashboard so that analytical results can be explored by business-oriented users.

3. Objectives

1. Develop a systematic data analytics pipeline for business datasets covering data collection, preprocessing, EDA, feature engineering and feature selection.

2. Analyze sales performance, customer behavior, profitability, regional performance and product-related business trends.

3. Develop and compare Decision Tree, Random Forest, XGBoost, LightGBM and CatBoost models.

4. Evaluate model performance using appropriate classification and regression metrics identified in the Project Approach Roadmap.

5. Apply SHAP and LIME to provide transparent explanations of machine-learning predictions.

6. Integrate analytical results and visualizations into an interactive Streamlit dashboard for business decision support.

7. Generate meaningful insights and actionable analytical outputs from the combined visualization and machine-learning workflow.

4. Datasets

Dataset	Role in Project
Superstore Sales Dataset	Retail transactional data containing customer, product, sales, profit, discount, quantity, shipping and regional information.
Global Superstore Dataset	Multinational retail transactional data containing customer, product, sales, profit, shipping, market and regional information.
Brazilian E-Commerce Public Dataset (Olist)	Real-world e-commerce data covering orders, customers, products, sellers, payments, deliveries and customer reviews.

5. Literature Survey

The following recent literature was selected around the project's main research themes: AI-assisted decision support, business analytics, explainable AI, machine-learning decision systems, and visualization for decision-making. The shortlist is intended for Review–1 literature coverage and should be retained alongside the final PDFs/DOIs used by the team.

Year	Authors	Paper	Source	Relevance
2024	Sokol & Flach	Interpretable representations in explainable AI: from theory to practice	Springer – Data Mining and Knowledge Discovery	Studies interpretable representations and their effect on explanation quality and reliability.

Year	Authors	Paper	Source	Relevance
2024	Bozorgpanah & Torra	Explainable machine learning models with privacy	Springer – Progress in Artificial Intelligence	Examines the trade-off between explainability and privacy, including TreeSHAP.
2024	■ ernevi■ien■ & Kabašinskas	Explainable artificial intelligence (XAI) in finance: a systematic literature review	Springer – Artificial Intelligence Review	Reviews XAI applications and identifies SHAP, feature importance and related methods as widely used.
2024	Brughmans, Melis & Martens	Disagreement amongst counterfactual explanations: how transparency can be misleading	Springer – TOP	Shows that different explanation methods can disagree, highlighting the need for careful explanation evaluation.
2024	Atzmueller et al.	Explainable and interpretable machine learning and data mining	Springer – Data Mining and Knowledge Discovery	Discusses explainability, interpretability, model transparency and their importance in data mining and decision support.
2024	Iovane & Chinnici	Decision Support System Driven by Thermo-Complexity: Scenario Analysis and Data Visualization	Applied Sciences	Connects dimensional analysis, scenario analysis and visualization with decision-support system design.
2024	Brumar et al.	A Typology of Decision-Making Tasks for Visualization	Visualization community research	Defines decision-making tasks for visualization and supports more systematic design of decision-support visualizations.
2024/2025	Haas & Hüllermeier	Conformalized prescriptive machine learning for uncertainty-aware automated decision making	Springer – International Journal of Data Science and Analytics	Demonstrates uncertainty-aware ML for business decision processes while retaining human decision control.
2025	Staudinger et al.	Knowledge graph support for descriptive business analytics	Springer – Decision	Addresses the difficulty of interpreting analytics results and retrieving the knowledge needed to justify business decisions.
2025	Lyu & Wu	Explainable Artificial Intelligence for Business and Economics: Methods, Applications and Challenges	Wiley – Expert Systems	Reviews XAI methods and applications relevant to business and economics, including transparency and practical challenges.

6. Research Gap Identification

Gap 1. Recent literature demonstrates strong progress in explainable machine learning, but explanation quality, reliability and user understanding remain active concerns.

Gap 2. Business analytics research increasingly addresses decision support, yet analytical knowledge and context may remain difficult for decision-makers to retrieve and interpret.

Gap 3. Visualization research shows that visual interfaces can support decision-making, but the decision tasks and the presentation of analytical explanations need to be designed systematically.

Gap 4. Many studies focus on an individual aspect—such as a model, an explanation method, a visualization technique, or a specific domain—rather than presenting a unified business analytics workflow.

Gap 5. The proposed project therefore focuses on integrating preprocessing, EDA, comparative tree-based ML, XAI and interactive business visualization into one end-to-end decision-support workflow across multiple business datasets.

7. Innovation / Novelty

1. An end-to-end business analytics workflow that connects descriptive analysis, predictive modelling, explainability and visualization.

2. Comparative evaluation of five tree-based learning approaches within the same business decision-support pipeline.

3. Integration of SHAP and LIME so that model outputs are accompanied by interpretable evidence rather than predictions alone.

4. Use of multiple benchmark business/e-commerce datasets to examine whether the workflow can support different business perspectives.

5. A Streamlit-based interactive dashboard that brings KPIs, business visualizations, model outputs and explanatory insights together for decision support.

8. Design & Methodology

Stage	Planned Work
Data Collection	Superstore Sales, Global Superstore and Olist datasets.
Data Preprocessing	Missing-value handling, duplicate removal, outlier detection/treatment, encoding, scaling/normalization and transformation.
Exploratory Data Analysis	Dataset overview, data quality assessment, descriptive statistics, univariate/bivariate/multivariate analysis and correlation analysis.
Business Analytics	Sales trend analysis, customer segmentation, profitability analysis, regional performance and product-category analysis.
Feature Preparation	Feature engineering followed by feature selection.
Machine Learning	Decision Tree, Random Forest, XGBoost, LightGBM and CatBoost.
Explainable AI	SHAP and LIME analyses for model interpretation.
Evaluation	Accuracy, Precision, Recall, F1-Score, ROC-AUC, MAE and RMSE as applicable to the selected modelling tasks.
Visualization & Dashboard	Matplotlib/Plotly visualizations and an interactive Streamlit dashboard.
Decision Support	Present analytical findings, model results and explanations as business-oriented insights.

9. Feasibility

Aspect	Assessment
Technical Feasibility	The project uses established Python-based data science and visualization technologies, including Scikit-learn, XGBoost, LightGBM, CatBoost, SHAP, LIME, Plotly, Matplotlib and Streamlit.
Data Feasibility	Three public/business benchmark datasets are identified in the Project Approach Roadmap, providing transactional and e-commerce information for analysis.
Implementation Feasibility	The proposed pipeline is modular: data processing, analysis, modelling, explainability and dashboard components can be developed and tested independently before integration.
Evaluation Feasibility	The roadmap already defines a set of classification and regression evaluation metrics, enabling quantitative model comparison.
Deployment Feasibility	The roadmap defines a deployment architecture with application, model, visualization and supporting infrastructure layers; deployment can be finalized after local validation.

10. Project Plan

Phase	Work	Output
Phase 1	Requirements, literature survey and research-gap analysis	Review–1 foundation
Phase 2	Dataset acquisition, validation and project environment setup	Reliable development base
Phase 3	Data cleaning, preprocessing and EDA	Prepared analytical datasets

Phase	Work	Output
Phase 4	Feature engineering and feature selection	Model-ready features
Phase 5	ML model development and comparative evaluation	Validated candidate models
Phase 6	SHAP/LIME explainability analysis	Interpretable model outputs
Phase 7	Business visualizations and Streamlit dashboard	Interactive analytics interface
Phase 8	Integration, testing and performance evaluation	Integrated decision-support prototype
Phase 9	Deployment, documentation, GitHub and final demonstration	Reproducible project deliverable

11. Expected Outcomes

1. A structured and reproducible business-data analytics pipeline.

2. EDA and visualization outputs describing sales, profitability, customers, products and regional trends.

3. Comparative results for the selected machine-learning models.

4. Explainable predictions supported by SHAP and LIME.

5. An interactive Streamlit dashboard integrating business KPIs, visual analytics and model-based insights.

6. A decision-support prototype that bridges analytical results and business-oriented interpretation.

12. References for Literature Survey

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